Symmetry Spending Under a Quantum Shot Budget: What It Takes to Certify That Shots Were Well Spent

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Abstract

A quantum predictor that averages a model response over the images of its input under a symmetry group is a *stochastic, shot-limited object*: it applies one of several circuits and reads out a finite number of measurement outcomes. Its risk therefore depends on how its budget was spent, and two allocations with the same total number of shots can have different risks — there is no ideal limit that both are approximating. We formalise this as a *risk-at-budget* functional and decompose it into three separately measurable parts: an orbit-mean model bias, a group-draw variance that more shots cannot reduce, and a shot variance that more group draws cannot reduce. The decomposition is exact and we verify it against brute-force enumeration to 3×10^{-17} rather than asserting it. Three things follow, and the third is the paper's finding.

First, at fixed total shots the split between "how many symmetry images" and "how many shots per image" leaves the shot term untouched, so orbit randomisation is free in shots. The interior optimum in that split, which an earlier version reported, is *retracted*: it came from charging eight distinct circuits in an experiment with two (§4.5).

Second, the calibration cost of a decision decomposes bound-independently as $\mathrm{Var}(D) = \mathrm{Var}_x(\mathbb{E}[D \mid x]) + \mathbb{E}[\mathrm{Var}(D \mid x)]$: deployed noise, which shots reduce, and between-instance variability of the contrast, which only more labelled preparations reduce, down to a floor no number of shots crosses. *Where* a certification frontier sits is a property of the inequality used, not of the problem, and every frontier number produced before that was understood is confined to a historical appendix (§8).

Third, and the reason for this submission: **a closed-form answer to when symmetry measurements repay their cost.** The mixture that spends wS shots across the orbit and $(1-w)S$ on raw has strictly convex risk, with optimum $w^*(S) = (\Delta_{\mathrm{ideal}} + A)/(2A) - V/(2AS)$, so symmetry is a luxury good, worth nothing below $S_{\mathrm{on}} = V/(\Delta_{\mathrm{ideal}} + A)$ and optimal only above $S_{\mathrm{full}} = V/(\Delta_{\mathrm{ideal}} - A)$. Estimated from shots, the rule beats both no symmetrisation and full averaging on a physical task where full averaging is worse than using no symmetry at all. Its allocation counterpart bounds what cleverness can recover: Neyman weighting of the orbit share recovers exactly $\mathrm{Var}_g(\sigma)$, which vanishes for any measure-preserving group action, so the cost is irreducible on every orbit this study contains and recoverable only when the action distorts the input law — both sides are measured.

Two earlier headline claims are withdrawn. The claim that the group-free arm is cheaper on six of seven tasks rested on an arm definition that scored the group arm's raw baseline with half its shots and on a frontier whose grid started where both arms already crossed; with a common-estimand record and a fixed total budget the comparison is *mixed*, with the group arm ahead on one physical task and the group-free arm ahead on another. The claim that a valid certificate is needed to select safely is also not supported: generic plug-in selection matches or beats the certified rule everywhere we looked, and the distribution-free rules never fire at these gaps.

We also report what a correct simultaneous distribution-free certificate costs here — five to twenty thousand labelled preparations for gaps of order 10^{-3}, a cost of that particular conservative bound and not a lower bound for the problem — and, on a bond-alternating Heisenberg response with a control-line rectification error, that the sign of orbit averaging's benefit is governed by how much odd signal the target contains rather than by whether a symmetry exists. Every number in this paper is emitted from the result files by scripts that also generate the sentences quoting them.

1 Introduction

A growing body of work asks how a fixed quantum measurement budget should be spent. Catak [1] derives a distribution-free bound separating a sample term from a shot term and a closed-form allocation $n^\star, S^\star$ under $B = nS$. Chen [2] certifies the *release* of a symmetry constraint during training, with simultaneous confidence bounds, a post-selection oracle inequality for adaptive candidate families, and grouped-measurement shot accounting. Chen [3] treats finite-measurement inference of the symmetry group itself. Wang et al. [4] and Cai et al. [5] give randomised symmetry implementations with explicit bias–sampling tradeoffs.

This paper is about a different object, and the difference is not cosmetic.

> For a deployed predictor that draws k symmetry images of its input and spends S shots across them, the risk is a functional of the allocation. Spending the same S differently does not change how well we *know* the risk; it changes *which* risk we have.

Three things follow, and all three are measured rather than asserted.

(1) A decomposition with two invariances. The deployed risk separates as $R_S(a) = D_a + V_G/k + \Sigma/S$, where D_a is an integrated squared discrepancy,. The shot term Σ/S does not depend on k, and the draw term V_G/k does not depend on S. At fixed total shots, moving shots between "more images" and "more shots per image" leaves one term exactly fixed and moves the other. There is no interior optimum in shots alone: $k = S$ weakly dominates. The interior optimum that practitioners actually observe comes from the cost of *distinct circuits*, which is a different resource and is where a switching cost c enters.

(2) Certification is where the cost is, and the group has to pay for it. Deciding whether symmetry spending helps requires a calibration experiment on independent labelled instances, and the deployed predictor's own shot noise enters the calibration contrast. That noise is reducible by shots; the between-instance variability of the contrast is not, and it sets a floor on $\mathrm{Var}(D)$ that more shots leave untouched. What we do *not* claim is an allocation rule: the location of the certification frontier moves by roughly $4\times$ in n depending on which concentration bound is used, and the direction of the preparations-versus-shots tradeoff at matched budget reverses between them (§9.3).

(3) The symmetry's value is a property of the target, not of the model. On the same physical mechanism, changing only how much odd-under-mirror signal the prediction target contains flips full orbit averaging from a 0.00800 risk reduction to a 0.00212 risk increase.

2 Access model, costs, and estimands

Access. Scalar single-copy shots: one labelled state preparation per calibration instance, one circuit-execution batch per prediction, computational-basis measurement of a diagonal observable. No two-copy access, no shared Pauli measurements, no classical shadows. Every baseline receives exactly this access. Where a published method needs more — a twirl–swap test, say — we say so and do not compare it against a weaker version of itself.

Costs are counted separately. `shots` (measurement repetitions), `preparations` (independently prepared labelled states), `circuits` (distinct compiled circuits, one per symmetry image touched), and `calibration_labels`. Deployment cost per prediction is $S + c \cdot$ (distinct circuits), with c a declared stress parameter. A method that receives free calibration or free state knowledge invalidates the comparison; the policy menu and the feasibility rules below exist to make that impossible to do by accident.

Feasibility is a property of a policy, not a convention. With `per`$= \lfloor S/k \rfloor$, a policy with $k > S$ spends k shots while claiming S. This study found that bug in its own menu after a first frontier run, and the apparent power gain at small S had been bought with four times the declared shots. Every policy now carries `shots_used` and a `feasible()` predicate, the menu is filtered by it, and the dropped entries are reported.

Estimands. Primary: $\Delta_S(a) = R_S(a) - R_S(a_0)$ at matched total shots, against the raw policy a_0. Operational primary: $N_{\mathrm{cert}}(\epsilon, \delta)$, the number of calibration preparations at which the certificate of §4 selects a policy with $\Pr[\Delta_S(a_{\mathrm{sel}}) \le \epsilon] \ge 1 - \delta$ and power $\ge 1 - \beta$ when a useful policy exists. Fallback to a_0 is a safe abstention, not a success.

3 Risk at budget

For input x and group element g, let $\mu_g(x) = \mathbb{E}[\text{shot} \mid g, x]$ and let $\sigma_g^2(x) = 1 - \mu_g(x)^2$ for a ± 1-valued readout. A policy draws k images, spends S shots total with S/k per draw and equal weights, and outputs $Z_a(x)$. Then, exactly,

$$R_S(a) \;=\; \underbrace{\mathbb{E}_x\big[(\bar{\mu}_a(x) - y)^2\big]}_{\text{integrated squared discrepancy}} \;+\; \underbrace{\mathbb{E}_x\big[\mathrm{Var}_g \mu_g(x)\big]/k}_{V_G/k} \;+\; \underbrace{\mathbb{E}_x\big[\mathbb{E}_g\, \sigma_g^2(x)\big]/S}_{\Sigma/S}, \tag{1}$$

where $\bar{\mu}_a$ is the orbit-mean response. The shot term deserves its invariance spelled out: $\sum_j (1/k)^2 \sigma_j^2/(S/k) = \bar{\sigma}^2/S$, independent of k.

Table 1: Equation (1) verified on the 2-qubit control. The residual column is the difference between the enumerated risk and the closed form; it is machine precision, not a fit.

4 Certification rule, and the two comparisons it enables

4.1 The rule, and what it is worth

Calibration uses n independent labelled instances; the independence unit is the instance, not the shot or the seed. For each instance we deploy the candidate and a_0 and record $D_i = (Z_a - Y_i)^2 - (Z_{a_0} - Y_i)^2$, with $D_i \in [-4, 4]$, and bound $\mathbb{E}[D]$ simultaneously over the fixed menu with Maurer-Pontil Corollary 5, $\ln = \log(2m/\delta)$ and range 8. This is an application of a known result to a fixed finite menu, not a new bound.

Its cost is worth stating as a number, because it is why the rest of this section is about something else. For the gaps this decision turns on, the range term alone requires five to twenty thousand labelled preparations before the bound can close, independently of how shots are allocated. That is a property of this conservative bound, *not* a lower bound for the statistical problem and not a measurement-complexity result: a sharper distribution-free inequality, or one exploiting the paired structure beyond the contrast range, would move it. What it establishes is that in this study the valid rule almost never fires, so a report of a certification frontier at n in the hundreds was not doing distribution-free certification.

Two selectors are carried side by side and named for what they are. The *certified* selector uses the rule above and has the finite-sample guarantee. The *plug-in* selector is generic paired calibration: it selects the candidate with the most negative paired mean loss and has no guarantee at all. A third, Studentised selector is reported only as descriptive simulation evidence; Wilson intervals on its simulated violation rate quantify Monte-Carlo uncertainty in these regimes, not coverage for a future task.

4.2 (a) What the group buys, with honest measurement sharing

Every deterministic policy of a finite-shot record costs nothing extra to evaluate. Shrinkage is a rescaling of the raw estimate, so eight shrinkage levels need *one* raw batch; an earlier version of this study resampled a batch per level and charged all eight, inflating the group-free comparator by the number of levels. Here one calibration record is s shots per distinct circuit per instance, and every policy in an arm is computed from that record at zero additional shot cost. The arms then differ only in the circuits their candidate sets need: one for the group-free arm, $|G|$ for the group arm. That is the group's honest measurement price.

Only policies whose deployed semantics match a shared record are admitted: the raw baseline and full equal-shot averaging at s shots per orbit point for the group arm, raw plus shrinkage for the group-free arm. Policies that allocate shots per *draw* rather than per orbit point would gain shots they do not pay for under sharing while being scored by their deployed risk, comparing an estimator against a different estimand; they are studied in the deployed-cost experiment instead.

Superseded. The comparison below was produced with the estimand defect recorded in A6.0: the group arm's raw baseline was scored from s identity shots while its reported risk difference used $2s$, so the

Regime	Policy	R_{64}	bias^2	V_G	Σ	residual
help	a0	0.03955	0.03428	0.00000	0.33686	4.2e-17
help	strat	0.00571	0.00040	0.00000	0.34004	1.0e-17
help	mix25	0.02476	0.01948	0.00000	0.33766	4.5e-17
help	mix50	0.01419	0.00890	0.00000	0.33845	2.6e-17
help	mix75	0.00784	0.00254	0.00000	0.33925	1.0e-17
help	k1	0.03949	0.00040	0.03377	0.34004	7.6e-17
help	k2	0.02260	0.00040	0.03377	0.34004	4.9e-17
help	k4	0.01416	0.00040	0.03377	0.34004	1.9e-17
help	k8	0.00993	0.00040	0.03377	0.34004	5.2e-18
help	k16	0.00782	0.00040	0.03377	0.34004	6.9e-18
help	k32	0.00677	0.00040	0.03377	0.34004	7.8e-18
help	k64	0.00624	0.00040	0.03377	0.34004	6.9e-18
hurt	a0	0.03813	0.03287	0.00000	0.33686	2.1e-17
hurt	strat	0.11438	0.10907	0.00000	0.34004	5.6e-17
hurt	mix25	0.05086	0.04558	0.00000	0.33766	3.5e-17
hurt	mix50	0.06781	0.06252	0.00000	0.33845	1.1e-16
hurt	mix75	0.08899	0.08369	0.00000	0.33925	4.2e-17
hurt	k1	0.14816	0.10907	0.03377	0.34004	1.1e-16
hurt	k2	0.13127	0.10907	0.03377	0.34004	5.6e-17
hurt	k4	0.12283	0.10907	0.03377	0.34004	1.1e-16
hurt	k8	0.11860	0.10907	0.03377	0.34004	2.8e-16
hurt	k16	0.11649	0.10907	0.03377	0.34004	1.2e-16
hurt	k32	0.11544	0.10907	0.03377	0.34004	9.7e-17
hurt	k64	0.11491	0.10907	0.03377	0.34004	1.4e-16
interior	a0	0.02071	0.01544	0.00000	0.33686	1.4e-17
interior	strat	0.01089	0.00558	0.00000	0.34004	3.5e-18
interior	mix25	0.01192	0.00665	0.00000	0.33766	2.6e-17
interior	mix50	0.00736	0.00207	0.00000	0.33845	6.9e-18
interior	mix75	0.00702	0.00171	0.00000	0.33925	1.6e-17
interior	k1	0.04467	0.00558	0.03377	0.34004	6.9e-18
interior	k2	0.02778	0.00558	0.03377	0.34004	4.2e-17
interior	k4	0.01934	0.00558	0.03377	0.34004	6.9e-18
interior	k8	0.01512	0.00558	0.03377	0.34004	2.3e-17
interior	k16	0.01300	0.00558	0.03377	0.34004	1.7e-17
interior	k32	0.01195	0.00558	0.03377	0.34004	3.5e-18
interior	k64	0.01142	0.00558	0.03377	0.34004	1.7e-17

Table 2: Risk-at-budget decomposition, enumerated exactly on the 2-qubit control. Residual is the gap between the enumerated risk and Eq. (1).

calibration contrast estimated a different quantity. It is retained here only to show what the defect did, and its conclusions are withdrawn in favour of §5.3.

Measured on `heisenberg_raw_n6` at $s = 8$ (so the scored difference used $S = 16$ shots): the deployed risk difference is $+0.00160$ while the calibration contrast measured -0.05097. The offset is -0.05258 against the predicted $\sigma_0^2/(2s) = -0.05258$, i.e. 32.8 times the effect under study.

Under shared measurement records and matched total resources, the group-free arm reaches the 0.9 power threshold at lower total cost than the group arm on 6 of 7 tasks, including both physical mechanisms.

The frozen A2.7/A3.5 stop gate, which requires the group arm to win on both physical mechanisms, therefore FAILS, and the scoped negative is the finding.

The valid Maurer-Pontil rule reaches 0.9 power somewhere in this sweep: yes, at small shots per circuit.

On the physical mechanisms the group-free arm is not merely cheaper but better: at matched shots per circuit, mild shrinkage of the raw estimate reduces deployed risk by more than full orbit averaging does on both the Heisenberg and the cluster task, as the two Δ_{best} columns of Table 3 show without needing a prose restatement. The group's bias reduction is real — it is the same effect measured in Table 10 — but at these budgets it is smaller than what the same shots buy when spent on the identity circuit alone.

Task	Δ_{best}(grp)	Δ_{best}(grp-free)	ϵ	grp cost	grp-free cost	verdict
cluster_even_n7	−0.00229	−0.01396	0.00386	17600	8800	group-free cheaper
cluster_raw_n7	−0.00229	−0.01396	0.00386	17600	8800	group-free cheaper
control_help	−0.03186	−0.00645	0.00000	17600	8800	group-free cheaper
control_hurt	+0.07367	+0.00650	0.00000	none	none	neither reaches
control_interior	−0.00884	−0.00363	0.00000	17600	8800	group-free cheaper
heisenberg_even_n6	−0.00854	−0.04795	0.00217	17600	8800	group-free cheaper
heisenberg_raw_n6	+0.00122	−0.04371	0.00217	17600	8800	group-free cheaper

Table 3: A3.4(a) and A3.5: matched-total-resource comparison with shared measurement records (one record per distinct circuit per instance; every deterministic policy evaluated from it at zero additional shot cost). Δ_{best} is quoted at a common $s = 8$ shots per circuit, because pooling across unequal deployments makes the arms incomparable. Costs are total shots including the declared deployment of 1000 predictions; the group arm necessarily pays for $|G|$ circuits.

4.3 (b) What certification buys, on the same candidate set

On the same symmetry menu, generic plug-in selection matches the Studentised selector wherever either reaches high power and dominates the certified rule, which almost never fires. This is the expected outcome and it was stated before running: the certificate in this paper *is* generic paired certification, so there is no symmetry-specific method advantage to find. Any advantage in §4.2 is therefore an advantage of a candidate set at a measurement price — never a better certification estimator, allocation rule, or shot complexity from exploiting group structure.

4.4 Resource ledger

Five quantities are counted separately. Preparations equal shots: the access model is scalar single-copy measurement, so every destructive shot consumes a freshly prepared state, and a batch of s shots at one circuit costs s preparations, not one. Compilations are the number of *distinct* circuits across a menu, $|G|$ for a group menu, not the sum over policies. Switches come from the schedule actually executed.

Task	Menu	n	S	total shots	power	violation
control_help	sym	200	8	20800	0.900 [0.85, 0.93]	0.000
control_help	generic	1600	8	123200	0.140 [0.10, 0.19]	0.000
control_hurt	sym	50	8	11200	0.000 [0.00, 0.02]	0.000
control_hurt	generic	50	8	11600	0.000 [0.00, 0.02]	0.000
control_interior	sym	200	32	96000	0.960 [0.92, 0.98]	0.000
control_interior	generic	1600	8	123200	0.100 [0.07, 0.15]	0.000

Table 4: A2.6 matched-total-cost comparison. Both menus are certified by the same procedure at the same declared ϵ; 'total shots' includes the factor of the menu size that calibration actually pays (A2.2). Rows show the cheapest allocation reaching power ≥ 0.9, or the best available.

4.5 Retraction: the switching optimum was an impossible circuit count

The control orbit has two elements, so at most two distinct circuits exist. The implementation reported $\min(k, 4|G|)$, reaching eight, and the switching experiment charged that number; the reported $k = 2$ optimum at large c was the price of circuits that cannot be compiled. With circuits correctly bounded, a per-circuit cost is flat for every $k \geq 2$ and the optimum is always at a boundary. Charging *switches* under an explicit interleaved schedule does produce an interior optimum, but in only 18 of 144 priced cells, and in none of the 72 per-circuit cells.

Charged resource	cells	interior optimum in k	max distinct circuits
per distinct circuit	72	0 (0.0%)	2
per switch	72	18 (25.0%)	2

Table 5: A2.3, the retracted switching result. With circuits correctly bounded by $|G| = 2$, a per-circuit price is flat for every $k \geq 2$ and never produces an interior optimum; the previously reported $k = 2$ optimum came from charging eight circuits. An interior optimum appears only under a per-switch price on an interleaved schedule.

5 When symmetry measurements repay their cost

The negative results above leave the question the study exists to answer. The exact risk functional (1) is enough to answer it in closed form for the mixture family, and the answer is budget-dependent.

Write the mixture-w predictor's mean as $\mu_0 + w(\bar{\mu} - \mu_0)$, where μ_0 is the raw response and $\bar{\mu}$ the orbit average, and set $A = \mathbb{E}[(\bar{\mu}-\mu_0)^2]$, $B = \mathbb{E}[(\mu_0-y)(\bar{\mu}-\mu_0)]$, $\Delta_{\text{ideal}} = \mathcal{D}_0-\mathcal{D}_G = -2B-A$, and $V = \bar{\sigma}^2-\sigma_0^2$, the shot-variance *cost* of symmetrisation. A mixture spending wS shots across the G orbit points and $(1-w)S$ on raw has estimator variance $[w\bar{\sigma}^2 + (1-w)\sigma_0^2]/S$, linear in w, and discrepancy $\mathcal{D}_0 + 2wB + w^2A$. Hence

$$R_S(w) = \mathcal{D}_0 + \frac{\sigma_0^2}{S} + 2wB + w^2A + \frac{wV}{S}, \qquad R_S''(w) = 2A > 0, \tag{2}$$

strictly convex with unique minimiser

$$w^*(S) = \frac{\Delta_{\text{ideal}} + A}{2A} - \frac{V}{2AS}. \tag{3}$$

Three consequences, each checked against the enumerated risk rather than asserted. Symmetry is a *luxury good* when $V > 0$: $dw^*/dS = V/(2AS^2) > 0$, and below $S_{\text{on}} = V/(\Delta_{\text{ideal}} + A)$ the optimal weight is exactly zero — no symmetry measurement is worth buying at that budget. Full symmetrisation is optimal only above $S_{\text{full}} = V/(\Delta_{\text{ideal}} - A)$, so partial symmetrisation occupies an explicit band, which is the pattern the physical-task risk tables showed empirically before this derivation existed. And when $V \leq 0$ symmetrisation also reduces shot variance, so it pays at every budget.

5.1 Does the rule work when the coefficients come from shots?

A, Δ_{ideal} and V are quadratic in orbit means, so plugging shot means in adds a positive noise bias. Each quadratic functional is therefore formed as a product of one statistic from each of two independent half-records: since the halves are independent and conditionally unbiased, $\mathbb{E}[(\bar{m}^A - m_0^A)(\bar{m}^B - m_0^B) = (\bar{\mu} - \mu_0)^2$ exactly. No bias correction and no unknown constants enter.

Every arm below spends exactly S shots per prediction and the same calibration shots, so calibration and deployment are matched on both sides — the mismatch that invalidated the earlier comparison.

In 11 of 18 cells the estimated w^* attains lower deployed risk than *both* no symmetrisation and full symmetrisation at the same shot budget; the cells are `control_hurt` ($S = 16$), `control_hurt` ($S = 64$), `control_hurt` ($S = 256$), `control_interior` ($S = 16$), `control_interior` ($S = 64$), `control_interior` ($S = 256$), and others.

The interior case is not a numerical curiosity: on `heisenberg_raw_n6` the ideal-predictor gain is negative, so full symmetrisation is worse than using no symmetry at all, and the theorem's partial weight beats both.

The estimator is noisy where shots are scarce: at $S = 16$ its standard deviation reaches 0.4 and regret against the best arm rises to 0.003, while by $S = 256$ it is 0.05 or below with regret under 2×10^{-4}. The theorem is therefore a rule for the regime where calibration can afford it, and the paper says so rather than presenting $\hat{w}$ as uniformly better.

5.2 What a smarter allocation can and cannot recover

Equation (3) prices symmetrisation at $V = \bar{\sigma}^2 - \sigma_0^2$ under uniform allocation over the orbit. Averaging with equal weights but allocating the orbit share by Neyman weights $n_g \propto \sigma_g$ instead charges $(\frac{1}{G}\sum_g \sigma_g)^2 - \sigma_0^2$,

Task	S	Δ_{ideal}	V	w^*	$\hat{w}$	$R(\hat{w})$	R(raw)	R(full)
control_help	16	+0.02917	+0.00577	0.99	0.97 ± 0.06	0.02281	0.05557	0.02198
control_help	64	+0.02917	+0.00577	1.00	0.98 ± 0.03	0.00568	0.03858	0.00561
control_help	256	+0.02917	+0.00577	1.00	0.99 ± 0.02	0.00177	0.03504	0.00177
control_hurt	16	−0.06939	+0.00577	0.00	0.00 ± 0.00	0.05475	0.05534	0.13083
control_hurt	64	−0.06939	+0.00577	0.00	0.00 ± 0.00	0.03791	0.03839	0.11516
control_hurt	256	−0.06939	+0.00577	0.00	0.00 ± 0.00	0.03400	0.03436	0.11034
control_interior	16	+0.00767	+0.00577	0.62	0.64 ± 0.15	0.02423	0.03633	0.02632
control_interior	64	+0.00767	+0.00577	0.63	0.64 ± 0.06	0.00688	0.02111	0.01091
control_interior	256	+0.00767	+0.00577	0.63	0.63 ± 0.03	0.00267	0.01690	0.00692
heisenberg_even_n6	16	+0.00676	+0.00076	0.98	0.66 ± 0.40	0.05482	0.06072	0.05407
heisenberg_even_n6	64	+0.00676	+0.00076	0.98	0.89 ± 0.16	0.01429	0.02219	0.01400
heisenberg_even_n6	256	+0.00676	+0.00076	0.98	0.97 ± 0.05	0.00407	0.01206	0.00410
heisenberg_raw_n6	16	−0.00164	+0.00076	0.38	0.53 ± 0.42	0.05546	0.05649	0.05752
heisenberg_raw_n6	64	−0.00164	+0.00076	0.38	0.37 ± 0.27	0.01607	0.01615	0.01885
heisenberg_raw_n6	256	−0.00164	+0.00076	0.38	0.37 ± 0.10	0.00563	0.00660	0.00887
cluster_n7	16	+0.00229	−0.00034	1.00	0.44 ± 0.46	0.04109	0.04314	0.03907
cluster_n7	64	+0.00229	−0.00034	1.00	0.77 ± 0.33	0.01068	0.01247	0.01009
cluster_n7	256	+0.00229	−0.00034	1.00	0.94 ± 0.10	0.00250	0.00507	0.00252

Table 6: A5: the closed-form symmetrisation weight w^*, estimated from calibration shots by cross-fitted products and deployed at exactly S shots per prediction, against the two extremes at the same cost. Δ_{ideal} is the ideal-predictor gain from symmetrisation and V its shot-variance cost; $\hat{w}$ is mean and standard deviation over 40 calibration repetitions.

and by Cauchy–Schwarz

$$\Big(\tfrac{1}{G}\sum_g \sigma_g\Big)^2 \;\le\; \tfrac{1}{G}\sum_g \sigma_g^2, \qquad \tfrac{1}{G}\sum_g \sigma_g^2 - \Big(\tfrac{1}{G}\sum_g \sigma_g\Big)^2 = \mathrm{Var}_g(\sigma), \tag{4}$$

so *the recoverable part of the symmetrisation cost is exactly the dispersion of the orbit's shot standard deviations*. Equal weights keep the estimator's mean at the orbit average, so the discrepancy is untouched and only the shot term moves: this is a pure structural statement about the measurement, not about the label.

That identifies when allocation can pay. Writing $\mu_g(x) = f(a_g x)$, if the action a_g preserves the input law then $\mu_g(X)$ and $\mu_0(X)$ are identically distributed, $\mathbb{E}[\sigma_g^2]$ is the same at every orbit point, and $\mathrm{Var}_g(\sigma) = 0$: on a measure-preserving orbit no allocation recovers anything, and the cost priced by V is irreducible. Every group action in this study is measure-preserving, which is why the effect is absent rather than small.

On the 5 tasks whose group action preserves the input distribution, $\mathrm{Var}_g(\sigma)$ is below 10^{-4} everywhere and Neyman allocation gains nothing: at $S = 256$ its gain is -0.000000, i.e. the integer rounding of the allocation, not a variance saving. On the one task where the action does not preserve the input law, $\mathrm{Var}_g(\sigma) = 0.014924$ and the gain is $+0.000058$ at $S = 256$, against the predicted $\mathrm{Var}_g(\sigma)/S = 0.000058$.

So the boundary is not a local null: the mechanism exists, its size is predicted exactly, and it vanishes for measure-preserving orbits, which is what every physical task in this study has.

This is the general boundary the previous version lacked. The mechanism is real and its size is predicted to three significant figures; it vanishes precisely for the orbits this problem class actually presents, which converts "we found no allocation gain" into a statement about when one could exist.

5.3 Equal-total-budget frontiers with parallel menus

The repaired comparison fixes the budget rather than the grid. The total B = calibration preparations + deployment preparations is held fixed, each arm chooses its instance count n, its per-prediction shot total S and its admissible policies subject to B, and the endpoint is *absolute realised deployed risk*. Realised risk is exact rather than simulated: the only randomness is the calibration record, which fixes the selection, and each candidate's $R_S(a)$ is known in closed form and verified against enumeration to 10^{-17}.

Task	σ_0	σ_1	$\mathrm{Var}_g(\sigma)$	S	gain = R(unif) − R(Ney)
control_help	0.5845	0.5874	0.000002	16	−0.000327
control_help	0.5845	0.5874	0.000002	256	−0.000000
control_interior	0.5845	0.5874	0.000002	16	−0.000327
control_interior	0.5845	0.5874	0.000002	256	−0.000000
heisenberg_even_n6	0.9125	0.9114	0.000000	16	−0.000817
heisenberg_even_n6	0.9125	0.9114	0.000000	256	−0.000000
heisenberg_raw_n6	0.9125	0.9114	0.000000	16	−0.000817
heisenberg_raw_n6	0.9125	0.9114	0.000000	256	−0.000000
cluster_n7	0.7966	0.7976	0.000000	16	−0.000624
cluster_n7	0.7966	0.7976	0.000000	256	−0.000000
control_asym_help	0.7044	0.4601	0.014924	16	+0.000778
control_asym_help	0.7044	0.4601	0.014924	256	+0.000058
control_asym_interior	0.7044	0.4601	0.014924	16	+0.000778
control_asym_interior	0.7044	0.4601	0.014924	256	+0.000058

Table 7: A6.5: the recoverable part of the symmetrisation cost is the orbit's shot-variance dispersion. Tasks whose group action preserves the input distribution have flat profiles and nothing to recover; the last two rows use the same label, model and group but an input law the action does not preserve, and the gain appears.

Two unfairnesses the referee's framing exposed are removed along with the estimand defect. The menus now carry **seven candidates each** from the same two families (unshrunk, and shrinkage toward the declared constant at five strengths), differing in exactly one structural respect — whether the base estimator uses the orbit. And shrinkage is available to *both* arms: giving it only to the group-free arm would have handed that arm a second variance-reduction device, the same unfairness class as the shot mismatch.

Task	Arm	$B = 4000$: risk	use.	f.b.	$B = 16000$: risk	use.	f.b.	$B = 64000$: risk	use.	f.b.
cluster_n7	group	0.17072	1.00	0.00	0.06536	1.00	0.00	0.01985	0.96	0.04
	group-free	0.17054	1.00	0.00	0.06629	1.00	0.00	0.02139	0.90	0.10
control_help	group	0.13394	1.00	0.00	0.04147	1.00	0.00	0.01067	1.00	0.00
	group-free	0.15307	1.00	0.00	0.06705	1.00	0.00	0.04255	0.84	0.15
control_hurt	group	0.19428	0.00	0.99	0.07222	0.00	1.00	0.04183	0.00	1.00
	group-free	0.17542	1.00	0.00	0.07222	0.00	1.00	0.04183	0.00	1.00
control_interior	group	0.13961	1.00	0.00	0.04703	0.96	0.04	0.01572	1.00	0.00
	group-free	0.13971	1.00	0.00	0.05145	1.00	0.00	0.02500	0.00	0.99
heisenberg_even_n6	group	0.14696	1.00	0.00	0.06547	1.00	0.00	0.02438	1.00	0.00
	group-free	0.12718	1.00	0.00	0.06826	1.00	0.00	0.03040	1.00	0.00
heisenberg_raw_n6	group	0.15154	1.00	0.00	0.07003	1.00	0.00	0.02937	0.58	0.41
	group-free	0.12753	1.00	0.00	0.06654	1.00	0.00	0.02638	1.00	0.00

Table 8: A6.3: best realised absolute deployed risk at a fixed total budget B, with the probability of useful selection and of falling back to the baseline. Each arm optimises n, S and its policy under the same B; both menus hold seven candidates at the same total S, and shrinkage is available to both arms. Lower is better.

At equal total budget the group arm attains lower realised risk in 10 of 18 (task, budget) cells and the group-free arm in 8. Violation probability is 0.000 in every cell for both arms.

Neither arm dominates, so neither a blanket advantage claim nor a blanket dominance claim is supported by this comparison.

Neither arm dominates. This replaces the earlier "cheaper on six of seven tasks", which was a property of how the arms were defined: that grid began where both arms already crossed, and the group arm's calibration was cheaper than its estimand required in one place and dearer in another. As A2.7 was written

it forbids an *advantage* claim, and it still does; it never licensed the stronger claim that symmetry spending is dominated, and that claim is withdrawn.

6 Physical tasks

Primary. Bond-alternating spin-1/2 Heisenberg chain, $n = 6$ and $n = 8$, open boundaries, quench from the Néel state, target the site-0 magnetisation. The control parameter treated as odd is the staggered field Δ, and the deployed policy pairs Δ with $-\Delta$.

What the symmetry claim requires, and where it fails. An earlier version asserted that $\Delta \to -\Delta$ *is* the chain mirror. It is not, and the check is cheap. The operator relation $MH(\lambda)M = H(-\lambda)$ does hold, to 8.9×10^{-16}. But a symmetry of the *deployed prediction* also needs the preparation and the readout to transform with it, and neither does: the Néel state maps to the shifted Néel state at distance $\sqrt{2}$, and Z_0 maps to Z_{n-1}. The relation that survives is $\langle Z_0\rangle_t(\lambda) = \langle Z_{n-1}\rangle_t(-\lambda)$, which holds to 10^{-15} and which describes a *different experiment* (mirrored preparation, mirrored readout). The relation this study actually exploits, $\langle Z_0\rangle_t(\lambda) = \langle Z_0\rangle_t(-\lambda)$, is off by 0.19. So $\Delta \to -\Delta$ is a paired control transformation whose odd component is measurable and cancellable, not a symmetry of the full experiment, and the risk decomposition is unaffected because it needs only the response functions μ_g, not a physical group action. The deployed device mis-sets that parameter by a second-order rectification of the analogue control line, $\Delta_{\text{eff}} = \Delta + c\Delta^2$, whose error is odd in Δ to leading order — which is exactly the component that averaging over $\Delta \to -\Delta$ cancels. A fixed-site readout bias would *not* be cancelable by input averaging, and we say so rather than claiming credit for a symmetry that the mechanism does not have.

The three regimes differ by how much of the odd response the target contains: $y_a = (1-a)\, y_{\text{even}} + a\, y_{\text{raw}}$, with y_{even} the part even under $\Delta \to -\Delta$. Table 10 shows raw, full averaging and the best partial mixture side by side, because a best-policy-only table cannot display a sign flip.

Task	$R(a_0)$	best policy	R(best)	Δ	max resid.
heisenberg_even_target_n6	0.02167	strat	0.01367	−0.00800	2.8e-17
heisenberg_mostly_even_n6	0.01870	mix75	0.01387	−0.00483	2.8e-17
heisenberg_raw_target_n6	0.01621	mix25	0.01525	−0.00096	2.4e-17
heisenberg_even_target_n8	0.01734	strat	0.01319	−0.00415	1.2e-17
cluster_even_target_n7	0.01137	strat	0.00934	−0.00203	1.6e-17
cluster_raw_target_n7	0.01137	strat	0.00934	−0.00203	1.6e-17

Table 9: Physical tasks at $S = 64$ total shots, enumerated risk. The sign of the benefit of orbit averaging flips with the odd content of the target.

Task	$R(a_0)$	full averaging	Δ	best mixture	Δ
heisenberg_even_target_n6	0.02167	0.01367	−0.00800	mix75: 0.01418	−0.00749
heisenberg_mostly_even_n6	0.01870	0.01425	−0.00445	mix75: 0.01387	−0.00483
heisenberg_raw_target_n6	0.01621	0.01833	+0.00212	mix25: 0.01525	−0.00096
heisenberg_even_target_n8	0.01734	0.01319	−0.00415	mix75: 0.01343	−0.00391
cluster_even_target_n7	0.01137	0.00934	−0.00203	mix75: 0.00946	−0.00192
cluster_raw_target_n7	0.01137	0.00934	−0.00203	mix75: 0.00946	−0.00192

Table 10: Raw, full averaging and the best partial mixture side by side (A2.5 and the reviewer's presentation point): the sign flip is visible only when all three are shown. Risks at $S = 64$ total shots, enumerated exactly.

Independent evaluation. A transverse-field three-body cluster Hamiltonian at $n = 7$, a different dynamical mechanism and a different observable, with the same rectification error. The split is by task family, declared before generation.

Quantity tested	residual
$\|MH(\lambda)M - H(-\lambda)\|$ (Hamiltonian only)	8.88e-16
$\|M\|\psi_0\rangle - \|\psi_0\rangle\|\|$ (Néel preparation)	1.41e+00
$\|MZ_0M - Z_0\|$ (site-0 readout)	2.00e+00
$\|\langle Z_0\rangle_t(\lambda) - \langle Z_{n-1}\rangle_t(-\lambda)\|$ (transformed experiment)	9.99e-16
$\|\langle Z_0\rangle_t(\lambda) - \langle Z_0\rangle_t(-\lambda)\|$ (fixed readout, as claimed)	1.94e-01

Table 11: A2.5: the mirror relation holds for the Hamiltonian and for the transformed experiment, and fails for the preparation and the readout actually used. Flipping only Δ is therefore a paired control transformation, not a symmetry of the deployed prediction.

7 What is and is not claimed

Claimed. (i) The risk-at-budget decomposition (1) with its two invariances, verified against an independent enumeration of the predictor and distinguished from the algebraic consistency check that an earlier version mislabelled as enumeration. (ii) That under a pure shot budget orbit randomisation is free in shots, and that the interior optimum in the split is *not* a circuit-count phenomenon (§4.5): it survives only as a per-switch cost on an interleaved schedule, in 18 of 144 priced cells. (iii) An equal-total-budget frontier with parallel seven-candidate menus in which neither arm dominates — the group arm is ahead on one physical task, the group-free arm on another — replacing an earlier "cheaper on six of seven" comparison that we show was built into its arm definitions, and in which generic plug-in selection is not beaten by the certified rule. (iv) The cost of a valid simultaneous distribution-free certificate at these effect sizes, stated as the cost of that bound and not of the problem. (v) On a physical mechanism, that the sign of orbit averaging's benefit tracks the odd content of the target, with the input flip explicitly *not* claimed to be a symmetry of the deployed experiment (§6).

Not claimed. No new concentration inequality. No joint sample/shot allocation rule for a generalization bound — that is Catak [1]. No certified symmetry release, no post-selection oracle inequality for adaptive families, no two-copy measurement primitive — that is Chen [2]. No symmetry discovery from finite probes — that is Chen [3]. **Withdrawn.** The interior switching optimum (§4.5); the claim that $\Delta \mapsto -\Delta$ is the chain mirror of the deployed experiment (§6, Table 11); the claim that the certification frontier locates a resource limit rather than a property of the inequality used (§9.3); and every coverage, power and frontier number produced by the pre-correction bound, which used $\log(4/(\delta n))$ inside a one-sided radius and the loss width where the contrast width belongs.

No device results, no quantum advantage, and no claim that stratified allocation $S_g \propto |w_g|\sigma_g$ is anything other than classical.

8 Limitations and what would falsify this

- Every task here is a dense statevector simulation with binomial shot sampling. The mechanisms are physical (control-line rectification, staggered-field mirror parity) but the data are not from hardware, and no device claim is made.
- The policy menu is finite and fixed before calibration. Policies generated from calibration losses would require the post-selection accounting of Chen [2], which this paper does not develop; our certificate would not cover them.
- The certification frontier depends strongly on the chosen inequality — by a factor of four in n between the two bounds we tried — and the direction of the preparations-versus-shots tradeoff at matched budget is not robust to that choice (§9.3). The bound-independent content is the variance decomposition, not a frontier location.
- On the independent-evaluation task the two target variants ($a = 0$ and $a = 1$) give identical risk to five decimals, i.e. that task's response is nearly even in Δ and does not exercise the odd-content axis at all. It is a transfer check for the decomposition, not for the sign-flip phenomenon.

- The independent-evaluation task is small ($n = 7$) and its rectification gap is an order of magnitude smaller than the primary task's, so it is a transfer check, not a demonstration.
- If a hardware or publicly measured dataset showed the opposite sign for the benefit of orbit averaging at matched shots, the decomposition would survive but the operational reading would not; that is the experiment that would most directly refute the useful part of this paper.

Historical (superseded) results

Nothing in this appendix supports any claim in this paper. It is retained because the objections raised in review are only checkable if the superseded numbers remain visible. These results were produced by the pre-correction bound (a radius with $\log(4/(\delta n))$ inside a one-sided confidence bound, and the loss width where the contrast width belongs) and by a ledger that counted one preparation per policy batch and summed circuits across policies. They are invalid for every certificate, coverage, power and frontier claim.

9 Certification, and what it costs

Calibration uses n independent labelled instances; the independence unit is the instance, not the shot or the seed. For each instance we deploy the candidate and a_0 and record $D_i = (Z_a - Y)^2 - (Z_{a_0} - Y)^2$, and we bound $\mathbb{E}[D]$ with an empirical-Bernstein inequality over the fixed menu, with the multiplicity split declared in advance. The inequality is deliberately ordinary: the contribution here is not a new concentration bound but the observation that what is being bounded is a difference of two *allocation-dependent* risks of stochastic predictors, selected with the same finite-shot data that defines them — precisely the independence assumption that Catak's bound requires and states [1].

n (preps)	S (shots)	violation	power	fallback
25	256	0.000	0.000	1.000
50	128	0.000	0.000	1.000
80	80	0.000	0.000	1.000
100	64	0.000	0.000	1.000
160	40	0.000	0.000	1.000
200	32	0.000	0.000	1.000
320	20	0.000	0.140	0.860
400	16	0.000	0.976	0.024
800	8	0.000	1.000	0.000
1600	4	0.000	1.000	0.000

Table 12: Certificate power along the preparations/shots axis at fixed total budget $B = nS = 6400$, averaging-helps regime, empirical-Bernstein bound, $\epsilon = 0$, $\delta = 0.05$.

Is the transition statistics, or the inequality? The empirical-Bernstein bound has a range term $7(b-a)\ln/(3(n-1))$ that is enormous at small n whatever the data says, so a frontier located at $n \approx 400$ could be a property of the bound rather than of the problem. We therefore report $\mathrm{Var}(D) = \mathrm{Var}_x(\mathbb{E}[D \mid x]) + \mathbb{E}[\mathrm{Var}(D \mid x)]$ separately from the bound's two terms. Only the second is reducible by shots; the first is reducible only by preparations. Table 13 gives both, and the split of the bound itself.

9.1 Matched total cost, including what calibration actually pays

The first version of this study labelled nS the calibration budget. It was not: the menu is evaluated on every calibration instance, so calibration costs $n\sum_p S_p$, a factor of the menu size larger, and preparations must be counted per shot-batch rather than per instance because repeated shots need repeated state copies. Table 14 repeats the comparison with the full five-part ledger at every point — labelled instances, preparations, shots, switches, compilations — and with a group-free comparator: the same certificate, the same calibration protocol, and a menu of shrinkage policies that buy variance reduction without touching the group.

Regime	S	Var(D)	between	noise	EB var. term	EB range term
help	4	0.08354	0.00507	0.07851	0.0000	-0.007
help	16	0.01186	0.00270	0.00917	0.0133	0.014
help	64	0.00414	0.00247	0.00168	0.0196	0.205
help	256	0.00281	0.00244	0.00038	nan	nan
hurt	4	0.13272	0.01042	0.12238	0.0000	-0.007
hurt	16	0.02866	0.00766	0.02104	0.0126	0.014
hurt	64	0.01197	0.00733	0.00466	0.0200	0.205
hurt	256	0.00837	0.00727	0.00113	nan	nan
interior	4	0.07145	0.00344	0.06805	0.0000	-0.007
interior	16	0.00751	0.00086	0.00665	0.0074	0.014
interior	64	0.00166	0.00060	0.00106	0.0152	0.205
interior	256	0.00078	0.00056	0.00023	nan	nan

Table 13: What the calibration contrast is made of, and what the bound adds. Var(D) splits into a between-instance part (only preparations reduce it) and a deployed-noise part (shots reduce it). The last two columns split the empirical-Bernstein bound at the matching allocation.

Task	Menu	n	S	total shots	power	violation
control_help	sym	200	8	20800	0.900 [0.85, 0.93]	0.000
control_help	generic	1600	8	123200	0.140 [0.10, 0.19]	0.000
control_hurt	sym	50	8	11200	0.000 [0.00, 0.02]	0.000
control_hurt	generic	50	8	11600	0.000 [0.00, 0.02]	0.000
control_interior	sym	200	32	96000	0.960 [0.92, 0.98]	0.000
control_interior	generic	1600	8	123200	0.100 [0.07, 0.15]	0.000

Table 14: A2.6 matched-total-cost comparison. Both menus are certified by the same procedure at the same declared ϵ; 'total shots' includes the factor of the menu size that calibration actually pays (A2.2). Rows show the cheapest allocation reaching power ≥ 0.9, or the best available.

The direction is structural rather than statistical. Shrinkage cannot produce a policy that beats raw by more than ϵ, because it moves the prediction toward a declared constant and therefore *adds* bias; only the group changes the bias term in Eq. (1). The symmetry menu reaches power 1.000 at 20,800 total shots in the helps regime and 1.000 at 96,000 in the interior regime; the group-free menu reaches 0.14 and 0.10 at comparable or greater cost and never certifies. In the hurts regime neither menu certifies, which is the correct behaviour: the certificate abstains. Across all 108 points the maximum observed violation rate is 0.015 against a nominal 0.05.

9.2 What a correct distribution-free certificate costs

With the corrected simultaneous bound the answer is analytic and unwelcome. The range term $7(b - a)\ln/(3(n-1))$ has $b - a = 8$ for a difference of squared losses and $\ln = \log(2m/\delta)$, so it decays as $1/n$ and, for the gaps this decision turns on ($|\Delta| \approx 0.008$), requires 5,109 to 20,679 labelled preparations before the bound can close — independently of how the shots are allocated. Distribution-free simultaneous certification of decisions this fine is not available at budgets this study can pay. The frontier numbers in this paper therefore use a Studentised bound whose coverage is audited rather than assumed (Table 15), and the distribution-free statement is the cost statement above.

9.3 Withdrawal: the frontier was mostly the inequality

An earlier version of this draft reported that certification is preparation-limited, on the evidence that at $B = 6400$ power jumped from 0.000 at $n \leq 200$ to 1.000 at $n \geq 800$. That claim is withdrawn. The empirical-Bernstein range term $7(b-a)\ln/(3(n-1))$ is ≈ 4 at $n = 25$ for a loss range of 4, so the bound, not the data, was deciding. Re-running the same experiment with a Studentised bound (Table 15) gives power

0.245/0.725/0.995/1.000 at $n = 25/50/100/200$ with $S = 128$, at a measured violation rate of 0.005 against a nominal 0.05 across all 90 cells — admissible empirically, though distribution-free validity is exactly what a Studentised bound does not offer, and a violation rate estimated on 400 repetitions per cell cannot resolve a true rate far below 0.05.

Bound	S	n=25	n=50	n=100	n=200	n=400
eb	8	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.93/0.00
eb	32	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	0.99/0.00
eb	128	0.00/0.00	0.00/0.00	0.00/0.00	0.00/0.00	1.00/0.00
normal	8	0.10/0.00	0.28/0.00	0.48/0.00	0.87/0.00	0.99/0.00
normal	32	0.42/0.00	0.92/0.00	1.00/0.00	1.00/0.00	1.00/0.00
normal	128	0.82/0.00	1.00/0.00	1.00/0.00	1.00/0.00	1.00/0.00

Table 15: Power/violation rate for the distribution-free empirical-Bernstein bound and the tighter Studentised bound, averaging-helps regime, $\epsilon = 0$, $\delta = 0.05$. A bound is admissible only where its measured violation rate stays at or below δ.

Reproducibility

Code: `shotcert/` (`core.py`, `qsim.py`, `tasks.py`, `physics.py`, `certify.py`, `experiments/`). Tests: `python3 -m unittest discover -s shotcert -t . -p 'test_*.py'` (35 tests, including the decomposition identity, the feasibility properties, and non-degeneracy checks for every task). Results: `shotcert/results/{control,physics,variance}.json`, compiled into `shotcert/RESULTS.md` by `shotcert/compile_results.py`; every number quoted here is emitted by that compiler. Protocol and novelty gate: `shotcert/docs/PROTOCOL.md`, `shotcert/docs/NOVELTY_MEMO.md`.

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